

URBAN AIR QUALITY ANALYSIS: UNDERSTANDING ENVIRONMENTAL DRIVERS AND PREDICTING AIR QUALITY

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DECLARATION

I hereby declare that this report, "Urban Air Quality Analysis: Understanding Environmental Drivers and Predicting Air Quality," is my own original work, completed as part of the IDRA Summer Training and Internship Program in Data Science and AI.

I personally carried out every stage of the project, including data preprocessing, exploratory data analysis, statistical analysis, feature engineering, and the implementation of the machine learning models, along with all the code that supports them. The analysis is based on the dataset provided through the IDRA training program, and the results, findings, and conclusions presented here come directly from that work.

Wherever I have drawn on external sources, datasets, or supporting materials, I have acknowledged them and cited them properly within the report. I have not knowingly copied or misrepresented anyone else's work, and I take full responsibility for the accuracy and integrity of the content presented in this report.

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ABSTRACT

Urban air pollution is a major environmental concern nowadays because of its effects on human health and the surrounding environment. This study has analyzed urban air-quality data to examine the variables, identify relationships between them, and build a machine learning model to predict the Air Quality Index (AQI). The dataset consists of 18,265 observations and 16 variables from five Indian cities, including measurements such as PM2.5, PM10, NO2, NOx, CO, SO2, O3, Benzene, Toluene, and Xylene. Before preprocessing, the data had been checked for missing values, duplicate records, inconsistent data types, and outliers. Exploratory and statistical analyses were then carried out to study the distributions of the variables and their relationships with AQI. The prediction task was treated as a regression problem with an 80:20 train-test split. Linear Regression and Random Forest Regression models were developed and were evaluated using MAE, MSE, RMSE, and R^2 . Correlations between the pollutant variables and AQI were found to be very weak, and both models performed poorly, with R^2 values close to zero. These results indicate that the available features had limited predictive power for AQI in this dataset. Hence this study highlights the importance of feature quality, additional environmental variables, and careful dataset construction in urban air-quality prediction.

Keywords: Urban Air Quality; Air Quality Index; Air Pollution; Data Science; Machine Learning

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1. INTRODUCTION

1.1 Background

Air pollution is a major environmental and public-health concern, particularly in urban areas, where emissions from transportation, industry, energy production, and construction can steadily degrade air quality. Monitoring and communicating air quality is therefore essential for understanding pollution levels and their potential effects on human health.

The Air Quality Index (AQI) is a tool used to communicate air-quality conditions to the general public in a simple and understandable form. It converts measurements of multiple air pollutants into a single index value along with a corresponding nomenclature and colour category. In India, the National Air Quality Index considers eight major pollutants: PM₁₀, PM_{2.5}, NO₂, SO₂, CO, O₃, NH₃, and Pb. The AQI is divided into six categories: Good, Satisfactory, Moderately Polluted, Poor, Very Poor, and Severe. These categories are determined using pollutant concentrations, sub-indices, and associated health breakpoints (Central Pollution Control Board, 2014).

The importance of maintaining good air quality extends beyond environmental monitoring because prolonged exposure to air pollution is associated with significant health risks. The United Nations General Assembly recognized the human right to a clean, healthy, and sustainable environment in 2022, highlighting the importance of a healthy environment for human well-being (United Nations General Assembly, 2022). The World Health Organization has also reported that ambient air pollution was associated with approximately 4.2 million premature deaths worldwide in 2019, demonstrating the substantial public-health burden associated with outdoor air pollution (World Health Organization, 2024).

These concerns highlight the need for systematic analysis of air-quality data to understand variations in pollution levels and AQI across urban areas. Data-driven techniques can help examine pollutant distributions, identify relationships among environmental variables, compare air-quality patterns across cities, and investigate whether the available variables contain sufficient information for predicting AQI. Therefore, analysing urban air-quality data provides an opportunity to combine statistical analysis, exploratory data analysis, and machine-learning techniques to better understand the patterns present within the available dataset.

1.2 Context of the Problem

In the real-world context, air pollution levels can vary considerably across cities due to differences in population density, transportation, industrial activities, and environmental conditions. These variations make continuous monitoring and analysis of urban air quality important.

Air-quality information is relevant to several stakeholders, including environmental agencies, public-health authorities, urban planners, researchers, and the general public. Environmental agencies can use air-quality data to monitor pollution levels, while

researchers and planners can analyse historical patterns to understand variations across different locations.

Data-driven analysis can further help identify relationships between pollutant concentrations and AQI and evaluate whether available environmental variables can be used to predict air-quality conditions. In this study, these aspects are examined using daily air-quality observations from Delhi, Mumbai, Chennai, Kolkata, and Bangalore, with the aim of understanding urban AQI patterns and evaluating the effectiveness of regression models for AQI prediction.

1.3 Scope of the Study

This study analyses daily air-quality observations from five Indian cities: Delhi, Mumbai, Chennai, Kolkata, and Bangalore. It covers data cleaning, exploratory and statistical analysis, and the evaluation of Linear Regression and Random Forest models for AQI prediction using MAE, MSE, RMSE, and R^2 . The aim is to assess whether the available pollutant and city variables contain enough information to predict AQI.

The study does not establish causal relationships between pollutants and AQI, nor does it function as a real-time monitoring system. Its conclusions apply only to the supplied dataset and the models tested.

2. PROBLEM STATEMENT AND MOTIVATION.

2.1 Problem Statement

Urban air pollution is influenced by multiple pollutants and environmental conditions, making it difficult to summarize and predict air-quality conditions using individual pollutant measurements alone. The Air Quality Index (AQI) provides a consolidated measure of air quality; however, understanding the factors associated with variations in AQI is important for data-driven monitoring and prediction.

The present study investigates the characteristics and relationships among major air pollutants and AQI across selected Indian cities. The study further develops a machine-learning regression model to estimate AQI from available pollutant measurements and city information. The objective is to determine whether the available variables contain sufficient predictive information to accurately estimate AQI and to identify important patterns within the dataset.

2.2 Motivation

The motivation for this study arises from the increasing importance of accessible and interpretable air-quality information in urban environments. Effective analysis of air-quality data can assist in identifying pollution patterns, comparing conditions across locations, and supporting data-driven environmental monitoring. By applying statistical analysis and machine-learning techniques, this project attempts to examine whether AQI can be predicted from the available pollutant-related variables

3. OBJECTIVES

- To understand the structure, characteristics, and quality of the selected urban air-quality dataset.
- To identify important statistical patterns and associations between the predictor variables and AQI.
- To build a machine learning model to predict AQI.
- To evaluate the model using appropriate performance metrics.
- To derive meaningful findings and recommendations from the analysis.

4. RESEARCH QUESTIONS

1. What are the major features and quality issues present in the dataset?
2. Which variables are the most strongly associated with the target outcome?
3. What relationships exist between the measured pollutants and AQI?
4. How accurately can the selected model predict the target outcome?
5. What are the limitations of the prediction model?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used in this study was provided as part of the India Data Research Academy (IDRA) Summer Training Program Capstone Project through the LMS. The dataset was used for educational and analytical purposes as provided within the training program.

5.2 Dataset Size and Structure.

The dataset contains 18,265 observations and 16 variables. The records represent daily air-quality observations for five cities: Delhi, Mumbai, Chennai, Kolkata, and Bangalore. The dataset contains pollutant measurements including PM2.5, PM10, NO, NO₂, NO_x, NH₃, CO, SO₂, O₃, Benzene, Toluene, and Xylene. It also contains the AQI value and corresponding AQI category.

Parameter	Description/Value
Dataset Name	P_8_city_day.csv
Total Number of Records	18,265
Total Number of Variables	16
Number of Cities	5
Cities Covered	Delhi, Mumbai, Chennai, Kolkata, Bangalore
Observation Frequency	Daily

Target Variable	AQI (Air Quality Index)
Numerical Variables	14
Categorical Variables	2
Date/Time Variable	Datetime
Missing Values	0
Duplicate Records	0
AQI Mean	251.11
AQI Median	251.20
AQI Standard Deviation	144.50
Minimum AQI	0
Maximum AQI	500

Table 1: Data Summary

The AQI variable was selected as the target because the primary objective of the study is to investigate and predict the numerical air-quality index. AQI_Bucket was not used as an input feature because it represents a categorical interpretation of AQI itself and would therefore introduce target leakage

5.3 Data Dictionary

Feature	Description	Type	Role
City	Name of the city	Categorical	Feature
Datetime	Date of observation	Date/Time	Feature
PM2.5	Fine particulate matter concentration	Numerical	Feature
PM10	Particulate matter concentration	Numerical	Feature
NO	Nitric oxide concentration	Numerical	Feature
NO2	Nitrogen dioxide concentration	Numerical	Feature
NOx	Oxides of nitrogen concentration	Numerical	Feature
NH3	Ammonia concentration	Numerical	Feature
CO	Carbon monoxide concentration	Numerical	Feature
SO2	Sulfur dioxide concentration	Numerical	Feature
O3	Ozone concentration	Numerical	Feature
Benzene	Benzene concentration	Numerical	Feature
Toluene	Toluene concentration	Numerical	Feature
Xylene	Xylene concentration	Numerical	Feature

AQI	Air Quality Index	Numerical	Target
AQI_Bucket	AQI Category	Categorical	Excluded from Regression

Table 2. Data Dictionary

6. DATA CLEANING

6.1 Missing Values

An inspection of the dataset identified no missing values across the 16 variables. All 18,265 observations contained complete records. Therefore, no rows were removed and no imputation technique was required.

6.2 Duplicate Records

The dataset contained zero duplicate records. Consequently, no observations were removed during duplicate-record handling.

6.3 Data Types and Inconsistent Values

The numerical pollutant and AQI variables were stored as numerical data types, while City and AQI_Bucket were categorical variables. Datetime was initially represented as an object/string field and was converted to a date-time format during preprocessing to allow temporal analysis.

No major inconsistent categorical values were identified in the City field.

6.4 Outlier Handling

The numerical variables were examined using the Interquartile Range (IQR) method:

$$\text{IQR} = Q3 - Q1$$

Values below:

$$Q1 - 1.5(\text{IQR})$$

or above:

$$Q3 + 1.5(\text{IQR})$$

were considered potential outliers.

The IQR analysis did not identify observations outside these limits for the numerical variables. Therefore, no observations were removed as statistical outliers.

This decision was taken to preserve the original observations rather than removing values solely because they appeared relatively high or low.

6.5 Before vs After Cleaning

Data Quality Measure	Before Cleaning	After Cleaning
Rows	18,265	18,265
Columns	16	16
Missing values	0	0
Duplicate records	0	0
IQR outliers detected	0	0

Table 3. Data Quality Before and After Cleaning

Overall, the dataset required limited conventional cleaning because it contained no missing values, duplicate records, or IQR-based outliers.

The comparison shows that the dataset retained all 18,265 observations and 16 variables after cleaning. No records were removed because there were no missing values, duplicate records, or observations identified as IQR-based statistical outliers. The main structural correction involved converting the Datetime field from an object/string representation into a date-time format for subsequent temporal analysis.

7. DATA PREPROCESSING

The cleaned dataset was prepared for analysis and machine learning by encoding categorical variables, scaling numerical features where required, selecting the relevant predictors, and dividing the data into training and testing sets. Preprocessing steps were implemented using a pipeline to ensure that information from the test set was not used during model training.

7.1 Categorical Encoding

The City variable was categorical and was converted into numerical representation using one-hot encoding. One-hot encoding was selected because the cities do not have a natural numerical order, and assigning numerical labels could incorrectly imply an ordered relationship.

The AQI_Bucket variable was excluded from the regression models because it is derived from AQI. Including it as a predictor could introduce target leakage and result in misleading model performance.

7.2 Feature Scaling and Transformation

Numerical features were standardized for the Linear Regression pipeline using standard scaling. Standardization transforms a variable according to:

$$z = \frac{x - \mu}{\sigma}$$

where x is the original value, μ is the training-set mean, and σ is the training-set standard deviation.

The scaling parameters were learned only from the training data and subsequently applied to the test data.

7.3 Train-Test Split

The dataset was divided into:

- 80% training data
- 20% testing data .

The training set was used to fit the model, while the test set was kept unseen during training and used for final evaluation.

7.4 Final Feature Set

The final predictive feature set consisted of:

City, PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, and Xylene

The target variable was:

AQI

Datetime and AQI_Bucket were excluded from the primary regression model.

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

```
plt.figure(figsize=(8, 5))

plt.hist(df["AQI"], bins=30, edgecolor="black")

plt.title("Distribution of AQI")
plt.xlabel("AQI")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

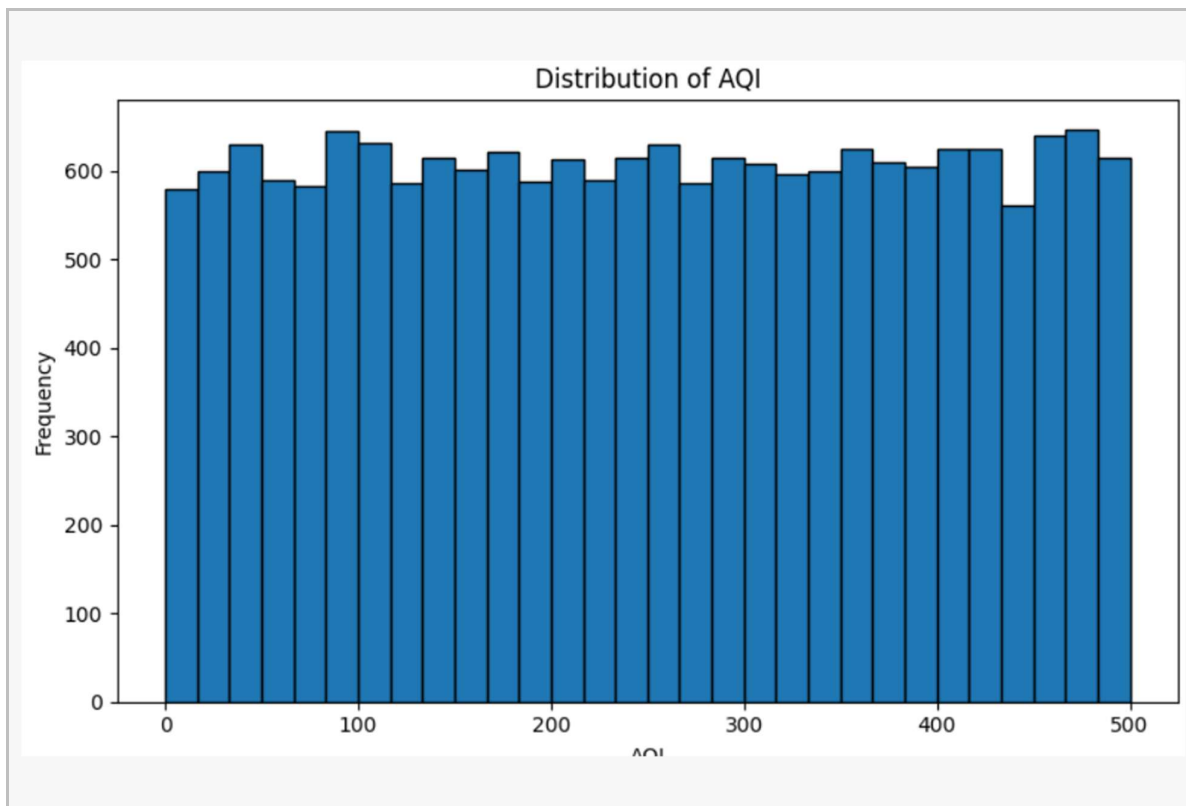


Figure 1. Distribution of [Important Variable]

The histogram shows the distribution of AQI values across the dataset. The AQI values cover a broad range, with a mean of approximately 251.11 and a median of approximately 251.20. The close proximity of the mean and median indicates that the distribution is approximately centered around 251..

8.2 Categorical Analysis

```
city_aqi = df.groupby("City")["AQI"].mean().sort_values(ascending=False)
plt.figure(figsize=(8, 5))
plt.bar(city_aqi.index, city_aqi.values)
plt.title("Average AQI by City")
plt.xlabel("City")
plt.ylabel("Average AQI")
plt.tight_layout()
plt.show()
```

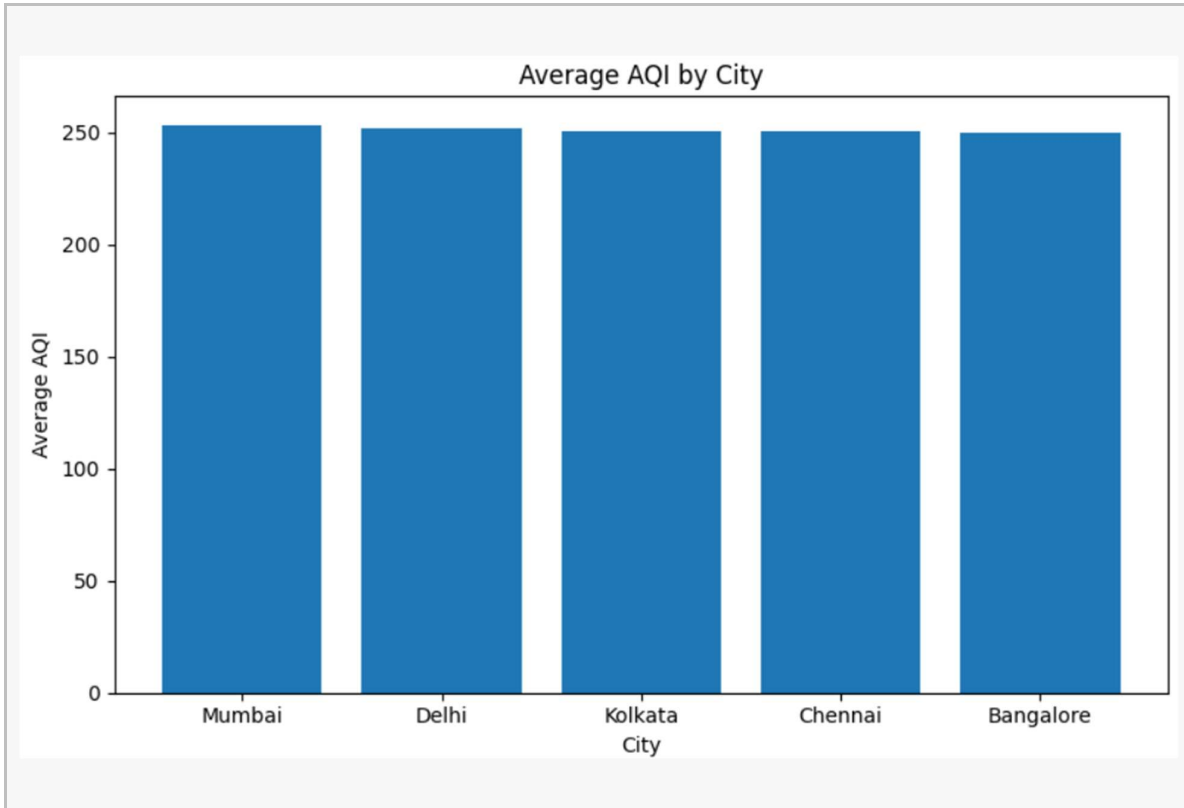


Figure 2. Average AQI by City

The bar chart compares the average AQI across the five cities in the dataset. Mumbai has the highest average AQI at approximately 253.32, while Bangalore has the lowest at approximately 249.77. The differences between the city averages are relatively small compared with the overall variation in AQI observed in the dataset.

8.3 Bivariate / Relationship Analysis

```
# Paste your relevant code here
plt.figure(figsize=(8, 5))
plt.scatter(df["PM2.5"], df["AQI"], alpha=0.25)
plt.title("Relationship Between PM2.5 and AQI")
plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.tight_layout()
plt.show()
```

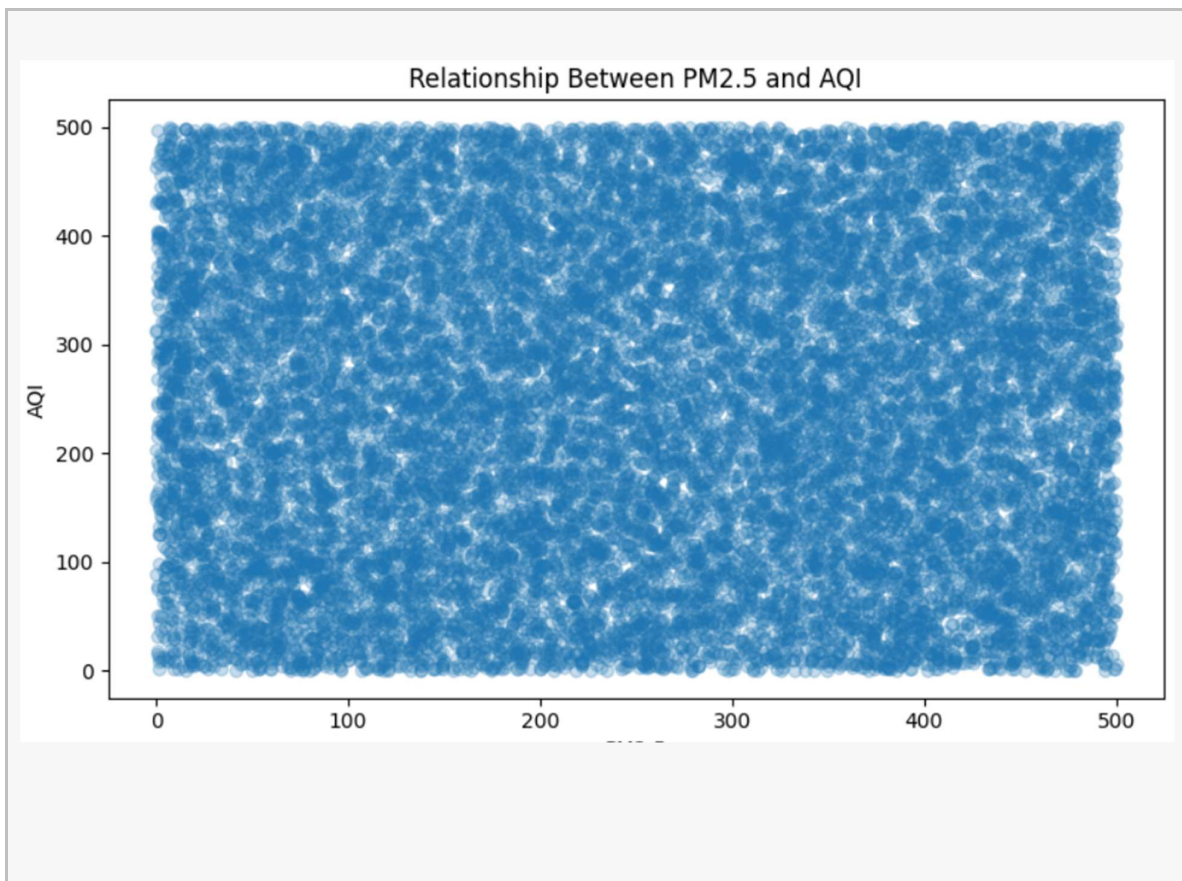


Figure 3. Relationship Between PM2.5 and AQI

The scatter plot was used to examine the relationship between PM2.5 concentration and AQI. The observations are widely dispersed across the plot, and no clear linear pattern is visible. This indicates that PM2.5 alone does not show a strong linear relationship with AQI within the supplied dataset. This observation is consistent with the very weak Pearson correlation observed between PM2.5 and AQI. However, the result describes the relationship within this dataset and does not imply that PM2.5 is unimportant to air quality in general.

8.4 Correlation Analysis

```
corr = df.select_dtypes(include=np.number).corr()
plt.figure(figsize=(12, 9))
sns.heatmap( corr, annot=True, fmt=".2f", cmap="coolwarm", center=0 )
plt.title("Correlation Heatmap of Numerical Variables")
plt.tight_layout()
plt.show()
```

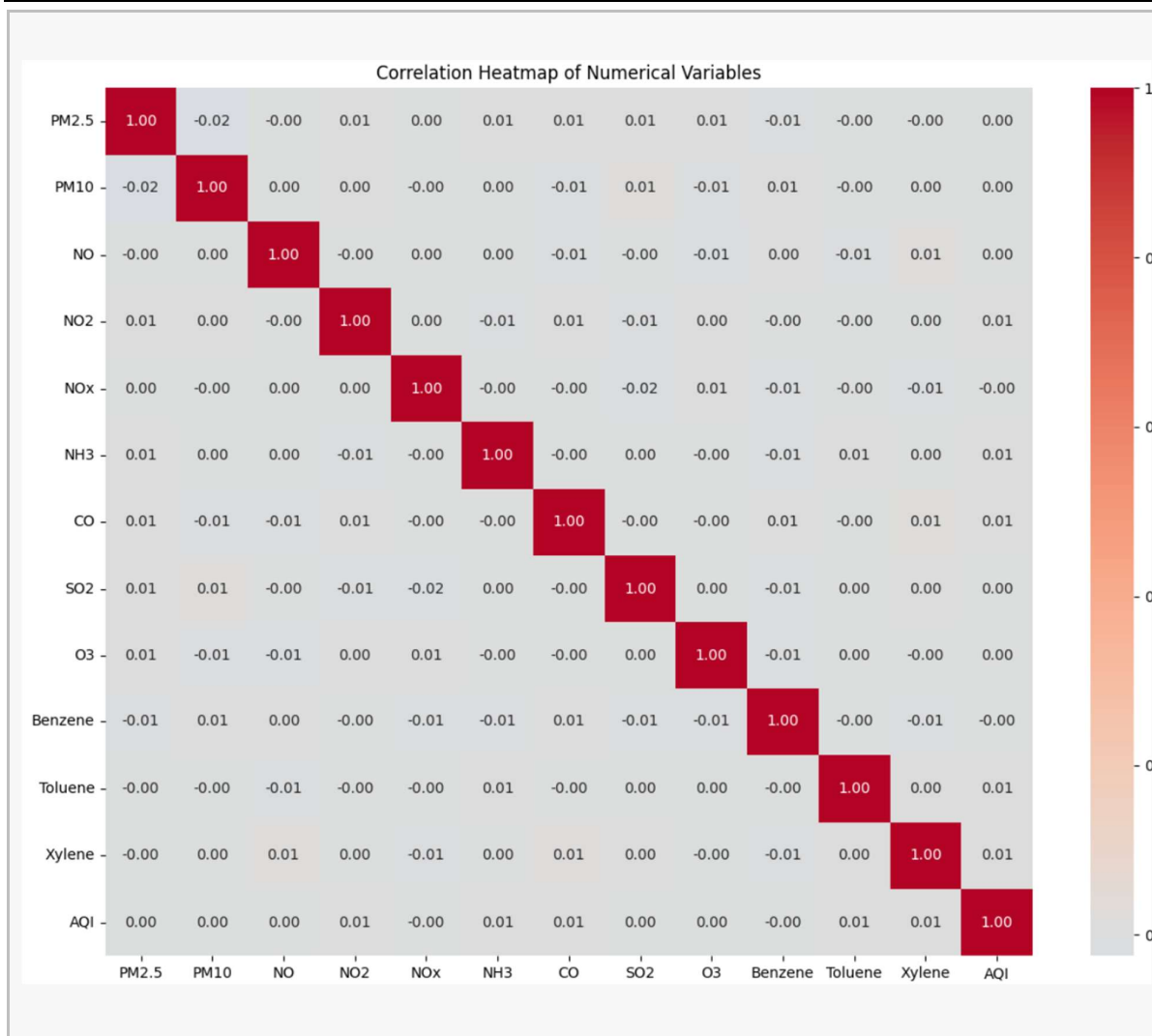


Figure 4. Correlation Heatmap of Numerical Variables.

The Pearson correlation analysis shows that the available pollutant variables have extremely weak linear correlations with AQI in the supplied dataset. The largest correlation with AQI is observed for NO₂, with a coefficient of approximately 0.0111, while the remaining pollutant variables also have coefficients very close to zero.

These results indicate that the available pollutant measurements do not demonstrate meaningful linear relationships with AQI within this particular dataset. This finding is relevant to the modelling stage because weak linear relationships may limit the predictive performance of a Linear Regression model. However, these results are specific to the supplied dataset and should not be interpreted as evidence that pollutants do not influence air quality in general.

8.5 Other Figures

```
df["Month"] = df["Datetime"].dt.month
monthly_aqi = df.groupby("Month")["AQI"].mean()
plt.figure(figsize=(9, 5))
plt.plot(monthly_aqi.index, monthly_aqi.values, marker="o")
plt.title("Average AQI by Month")
plt.xlabel("Month")
plt.ylabel("Average AQI")
plt.xticks(range(1, 13))
plt.tight_layout()
plt.show()
```

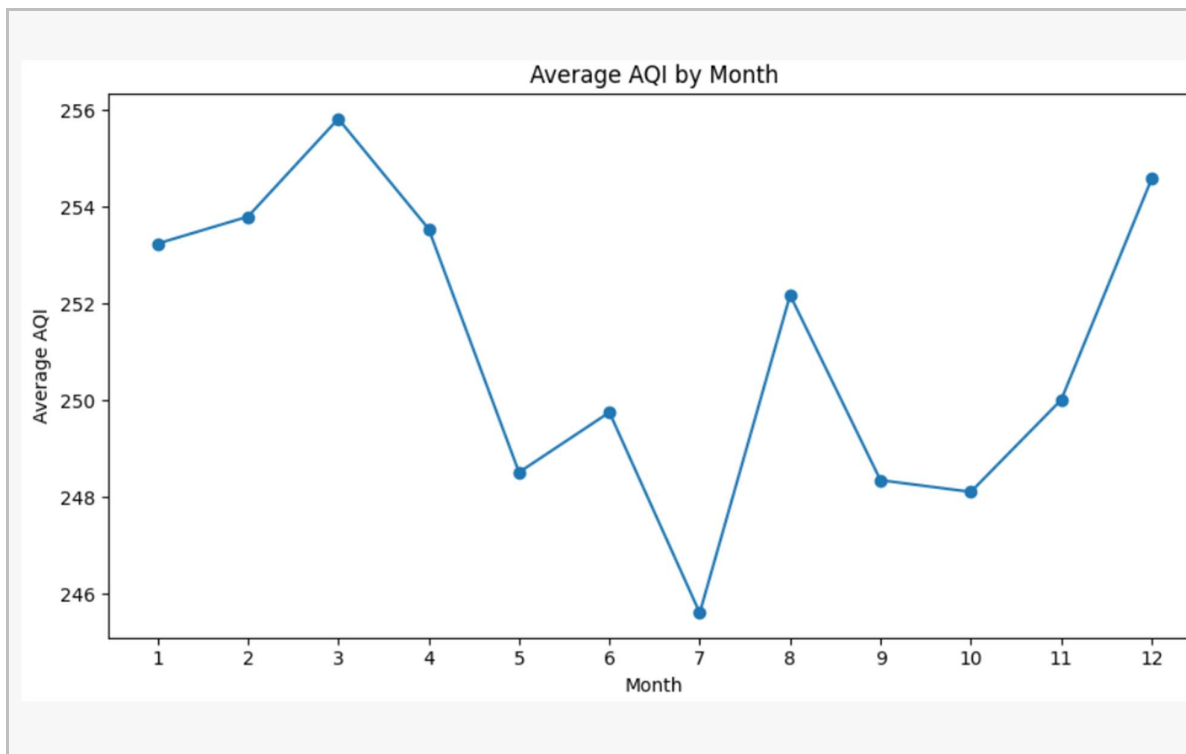


Figure 4. Average AQI by Month

The monthly AQI analysis was performed to examine whether the average AQI varied across the months represented in the dataset. The plot provides a temporal summary of AQI and helps identify month-to-month variation in the observations. The observed pattern should be interpreted as a descriptive temporal pattern rather than evidence that month itself causes changes in air quality. More detailed seasonal analysis would require additional environmental variables such as temperature, humidity, wind speed, rainfall, and atmospheric pressure.

9. STATISTICAL ANALYSIS

Variable	Mean	Median	Std. Dev.	Min.	Max.
PM2.5	250.60	251.00	144.46	0.00	499.90
PM10	299.44	300.30	173.48	0.00	600.00
NO2	75.42	76.00	43.46	0.00	150.00
CO	5.00	5.00	2.89	0.00	10.00
SO2	49.84	49.90	28.99	0.00	100.00
O3	100.41	100.70	57.59	0.00	200.00
AQI	251.11	251.20	144.50	0.00	500.00

Table 4. Descriptive Statistics of Key Variables.

Figure 4 shows that AQI has a relatively large standard deviation of approximately 144.50, indicating considerable variation in air-quality index values. The mean and median values for several variables are relatively close, suggesting that their distributions are broadly centered around their respective central values.

10. FEATURE ENGINEERING.

Original Variable	Engineered Variable	Transformation	Purpose
Datetime	Year	Extract year	Temporal analysis
Datetime	Month	Extract month	Monthly/seasonal analysis
Datetime	DayOfWeek	Extract day of week	Weekly temporal analysis

Table 5. Engineered Features

The Datetime variable was converted into a proper date-time representation and used to derive Year, Month, and DayOfWeek variables. These variables support temporal exploratory analysis by allowing observations to be grouped according to year, month, and day of the week.

For the primary AQI regression experiment, these derived temporal variables were not included as predictive inputs. The main model instead used City and the available pollutant measurements. This approach keeps the primary experiment focused on the environmental measurements supplied in the dataset.

No additional mathematical pollutant-combination features were introduced because the dataset already contained the measured pollutant variables required for the primary analysis.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The input features include measurements such as PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene and Xylene as well, as city information. The target variable is the AQI, which's a numerical value representing air quality.

This problem is formulated as regression because the AQI is a number that can take on a wide range of values. It is not a label so classification would not be appropriate. The goal is to predict the AQI value based on environmental data, which fits the definition of a regression problem. Predicting AQI as a number allows for more detailed and nuanced understanding of air quality levels.

11.2 Model Selection.

Two regression approaches were considered:

1. Linear Regression
2. Random Forest Regression

Linear Regression was chosen as the baseline model because it is simple and easy to understand. It lets us check if the pollutant variables we have linearly relate to AQI. Random Forest Regression was chosen as a backup. It is more flexible. Can find complex patterns and interactions between the features that Linear Regression might miss. Using both models lets us see how a simple method compares to a flexible one and to see if adding complexity helps predict AQI better. I evaluated both models with MAE, MSE, RMSE and R^2 , on the test set.

11.3 Training the Model

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
```

The model training was done by splitting the dataset into training and testing sets using an 80:20 ratio with set to 42. All data preparation steps were learned from the training data. Then applied to the test data. The City column was transformed using one-encoding. The numerical pollutant variables were standardized.

Two regression models were trained. Linear Regression was chosen as the baseline model because it gives an intuitive way to examine the relationship between the input variables and the target variable. We used Random Forest Regression as a model which is capable of capturing complicated patterns and interactions between features.

The Random Forest model was set up with 100 decision trees, `random_state` set to 42 and used processing, with `n_jobs` set to -1. The test dataset was never seen by the model during training. It was used for the final evaluation.

12. MODEL EVALUATION

12.1 Regression Evaluation

Model	MAE	MSE	RMSE	R ²
Linear Regression	124.63	20,828.16	144.32	-0.0009
Random Forest Regression	125.46	21,254.46	145.79	-0.0214

Table 6. Regression Model Performance

The models were evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R² on the unseen test set.

Linear Regression produced an MAE of approximately 124.63 AQI units and an RMSE of approximately 144.32 AQI units. Its R² value was approximately -0.001, indicating that the model explained essentially none of the variance in the test-set AQI relative to a simple baseline.

Random Forest Regression produced an MAE of approximately 125.46 AQI units and an RMSE of approximately 145.79 AQI units. Its R² value was approximately -0.021, which also indicates very limited predictive performance.

The two models therefore produced broadly similar results. The more complex Random Forest approach did not provide a meaningful improvement over the Linear Regression baseline. These results are consistent with the correlation analysis, which showed that the available pollutant variables had extremely weak linear associations with AQI in the supplied dataset.

12.2 Overfitting and Underfitting

Model	Train R ²	Test R ²
Linear Regression	0.0005	-0.0009
Random Forest Regression	0.8579	-0.0214

Table 6. Regression Model Performance

The training and testing models were compared using the R² values and how they would perform to unseen observations. The Linear Regression model produced a training R² of 0.0005 and a test R² of -0.0009. The similarity between these values indicates that the model was unable to capture substantial predictive relationships in both the training and testing data.

In contrast, the Random Forest model achieved a training R^2 of 0.8579 but a test R^2 of -0.0214 . The large difference between training and testing performance indicates that the Random Forest model fitted the training data strongly but failed to predict accurately to the unseen test observations. This is hence consistent with overfitting.

The results therefore suggest two different issues. First being that Linear Regression showed very limited ability to learn predictive relationships from the available features, while Random Forest was able to fit patterns in the training data. The negative test R^2 values for both models indicate that neither model was able to predict AQI on the unseen data.

13. RESULTS AND FINDINGS

1. Dataset Quality:

The dataset contained 18,265 observations across 16 variables and had no missing values or duplicate records. This indicates that the dataset was structurally complete and required no major corrections during the initial data quality assessment.

2. AQI Variation:

AQI values demonstrated substantial variation across the dataset, with a mean of 251.11 and a standard deviation of 144.50. This indicates considerable dispersion in AQI values among the recorded observations.

3. Weak Predictor Relationships:

Correlation analysis revealed that extremely weak linear relationships between the available pollutant variables and AQI. NO_2 showed the strongest correlation at approximately 0.011, which remains effectively close to zero within the supplied dataset.

4. Limited Predictive Performance

Neither model demonstrated reliable predictive performance on unseen data. Linear Regression achieved a test R^2 of -0.0009 , while Random Forest achieved -0.0214 , with the other model also displaying evidence of overfitting from its substantially higher training R^2 of 0.8579.

14. DISCUSSION

The analysis demonstrated an important distinction between possessing pollutant measurements and possessing sufficient predictive information for AQI modelling. Although the dataset contains measurements for pollutants that matter for air-quality analysis the statistical analysis found very weak linear ties between those variables and AQI.

The modelling results were consistent with this observation. Linear Regression produced a training R^2 of only 0.0005 and a test R^2 of -0.0009 , showing that it captured very little

predictive information from the available features. Random Forest achieved a training R^2 of 0.8579 but a test R^2 of -0.0214 . The large difference between its training and testing performance indicates that the model fitted the training observations strongly but failed to generalize to unseen observations.

Therefore, increasing model complexity did not solve the underlying prediction problem. Instead, the results suggest that the dataset features did not contain sufficient parameters for efficient AQI prediction.

These findings should be seen as traits of the supplied dataset not as proof that pollutants have no relationship with air quality. The limited performance may come from how the supplied data is structured and represented. Additional environmental variables, temporal information, measurement context, and a dataset with clearly documented AQI construction could provide a stronger basis for future prediction models.

15. LIMITATIONS

1. **Limited feature representation:** The dataset primarily focused on pollutant concentrations and city information and lacked important meteorological variables such as temperature, humidity, wind speed, atmospheric pressure, and rainfall.
2. **Weak predictive relationships:** The pollutant variables exhibited extremely weak correlations with AQI in the supplied dataset, limiting the effectiveness of conventional regression approaches.
3. **Model limitations:** Only a limited number of regression approaches were evaluated. More advanced models and validation techniques could potentially provide more additional insights.
4. **Dataset-specific conclusions:** The findings presented are specific to the supplied dataset and should not be generalized to all Indian cities or real-world air-quality monitoring systems.
5. **Temporal modelling limitations:** Although the dataset contains a date variable, the primary prediction experiment did not use a dedicated time-series forecasting methodology.

16. CONCLUSION

This study investigated urban air-quality data with the objective of understanding pollutant characteristics and developing a machine-learning model for AQI prediction. The dataset consisted of 18,265 observations across five cities and included measurements of multiple atmospheric pollutants alongside AQI.

The data-quality assessment showed that the dataset contained no missing values or duplicated records, while the IQR analysis did not identify any statistical outliers requiring removal. Exploratory and statistical analysis displayed substantial variation in AQI but extremely weak correlations between AQI and the available pollutant variables.

Two regression approaches, Linear Regression and Random Forest Regression, were evaluated. Both models demonstrated limited predictive performance, with R^2 values close to or below zero. The results therefore indicate that the supplied variables, in their current form, are insufficient for accurately predicting AQI using the tested approaches.

Despite the limited predictive performance, the project demonstrates a complete data-science workflow involving data understanding, cleaning, preprocessing, exploratory analysis, statistical analysis, feature preparation, machine-learning development, and evaluation. The findings also highlight the importance of dataset quality and feature relevance when developing environmental prediction models.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

Based on the analysis, the following recommendations are proposed:

- Inclusion of meteorological variables: Future datasets should incorporate factors like temperature, humidity, wind speed, atmospheric pressure, rainfall, and other relevant environmental variables.
- Strengthening of temporal analysis: Daily and seasonal patterns should be investigated since air-quality conditions can vary substantially over time.
- Improve data quality and feature relevance: Future modelling should use datasets in which pollutant measurements and AQI values preserve meaningful relationships based on a clearly documented AQI calculation methodology.
- Expanding geographic coverage: Including additional Indian cities and longer observation periods would drastically improve the scope of the analysis and potentially increase the validity of the findings.

17.2 Future Work

Future research can significantly build on this study by incorporating several diverse geographical factors—such as topography, land use, and urban density, alongside standard pollutant levels to capture a more holistic view of the atmospheric variables. Rather than estimating the Air Quality Index (AQI) for a few random, isolated data points, it would be highly valuable to deploy advanced time-series forecasting methods like LSTM or ARIMA to predict continuous future AQI values. Finally, ensuring the AQI calculation method is transparently documented will allow for highly reliable, actionable conclusions regarding the precise environmental drivers of urban air quality, ultimately aiding policymakers in developing targeted pollution-mitigation strategies.

18. REFERENCES

- 01) Central Pollution Control Board. (2014). National Air Quality Index. Ministry of Environment, Forest and Climate Change, Government of India.
<https://cpcb.gov.in/National-Air-Quality-Index/>
- 02) United Nations General Assembly. (2022). The human right to a clean, healthy and sustainable environment. Resolution A/RES/76/300. United Nations.
<https://digitallibrary.un.org/record/3983329?v=pdf>
- 03) World Health Organization. (2024). Sustainable development goal indicator 3.9.1: Mortality attributed to air pollution. World Health Organization.
<https://www.who.int/publications/i/item/9789240099142?>
- 04) World Health Organization. (n.d.). Air quality, energy and health: Exposure to air pollution. World Health Organization. <https://www.who.int/teams/environment-climate-change-and-health/air-quality-energy-and-health/health-impacts/exposure-air-pollution?>
- 05) World Health Organization. (n.d.). Guidance on air pollution and health. Compendium of WHO and other UN guidance on health and environment. World Health Organization. <https://www.who.int/tools/compendium-on-health-and-environment/air-pollution-and-health>
- 06) scikit-learn developers. (n.d.). LinearRegression. scikit-learn documentation. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html?
- 07) scikit-learn developers. (n.d.). RandomForestRegressor. scikit-learn documentation. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>
- 08) India Data Research Academy. (2026). P_8_city_day.csv [Dataset]. IDRA Summer Training Program, Data Science and AI.

APPENDIX A. CODE

A.1 Data Loading and Inspection

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

df = pd.read_csv("P_8_city_day.csv")

print("Dataset Shape:", df.shape)
print(df.head())

print("Missing Values:")
print(df.isnull().sum())

print("Total Missing Values:", df.isnull().sum().sum())

print("Duplicate Records:", df.duplicated().sum())

print("Before Conversion:")
print(df.dtypes)

df["Datetime"] = pd.to_datetime(df["Datetime"])

print("\nAfter Conversion:")
print(df.dtypes)
```

A.2 Data Cleaning

```
num_cols = df.select_dtypes(include=np.number).columns

outlier_summary = []

for col in num_cols:
    q1 = df[col].quantile(0.25)
    q3 = df[col].quantile(0.75)
    iqr = q3 - q1

    lower = q1 - 1.5 * iqr
    upper = q3 + 1.5 * iqr

    count = ((df[col] < lower) | (df[col] > upper)).sum()

    outlier_summary.append(
        [col, q1, q3, iqr, lower, upper, count]
    )
```

```

outlier_df = pd.DataFrame(
    outlier_summary,
    columns=[
        "Variable", "Q1", "Q3", "IQR",
        "Lower Bound", "Upper Bound", "Outliers"
    ]
)

display(outlier_df)

```

A.3 Preprocessing

```

X = df.drop(columns=["AQI", "AQI_Bucket", "Datetime"])
y = df["AQI"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

categorical_features = ["City"]

numerical_features = [
    "PM2.5", "PM10", "NO", "NO2", "NOx",
    "NH3", "CO", "SO2", "O3",
    "Benzene", "Toluene", "Xylene"
]

preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numerical_features),
        ("cat", OneHotEncoder(handle_unknown="ignore"),
         categorical_features)
    ]
)

```

A.4 EDA and Visualization

```

plt.figure(figsize=(8, 5))

plt.hist(df["AQI"], bins=30, edgecolor="black")

plt.title("Distribution of AQI")
plt.xlabel("AQI")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

#AVERAGE AQI BY CITY
city_aqi = df.groupby("City")["AQI"].mean().sort_values(
    ascending=False
)

plt.figure(figsize=(8, 5))

plt.bar(city_aqi.index, city_aqi.values)

plt.title("Average AQI by City")
plt.xlabel("City")
plt.ylabel("Average AQI")

```

```
plt.tight_layout()
plt.show()

#PM2.5 AND AQI RELATIONSHIP
plt.figure(figsize=(8, 5))

plt.scatter(
    df["PM2.5"],
    df["AQI"],
    alpha=0.25
)

plt.title("Relationship Between PM2.5 and AQI")
plt.xlabel("PM2.5")
plt.ylabel("AQI")

plt.tight_layout()
plt.show()

#CORRELATION HEATMAP
corr = df.select_dtypes(include=np.number).corr()

plt.figure(figsize=(12, 9))

sns.heatmap(
    corr,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0
)

plt.title("Correlation Heatmap of Numerical Variables")
plt.tight_layout()
plt.show()

#MONTHLY AQI TREND
df["Month"] = df["Datetime"].dt.month

monthly_aqi = df.groupby("Month")["AQI"].mean()

plt.figure(figsize=(9, 5))

plt.plot(
    monthly_aqi.index,
    monthly_aqi.values,
    marker="o"
)

plt.title("Average AQI by Month")
plt.xlabel("Month")
plt.ylabel("Average AQI")
plt.xticks(range(1, 13))

plt.tight_layout()
plt.show()
```

A.5 Model Training and Evaluation

```
#LINEAR REGRESSION
lr_model = Pipeline(
    steps=[
```

```
        ("preprocessor", preprocessor),
        ("model", LinearRegression())
    ]
)

lr_model.fit(X_train, y_train)

lr_pred = lr_model.predict(X_test)

print("Linear Regression trained.")

#RANDOM FOREST REGRESSION
rf_model = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        ("model", RandomForestRegressor(
            n_estimators=100,
            random_state=42,
            n_jobs=-1
        ))
    ]
)

rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)

print("Random Forest trained.")

#MODEL EVALUATION
def evaluate_model(name, y_true, y_pred):
    mae = mean_absolute_error(y_true, y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_true, y_pred)

    return [name, mae, mse, rmse, r2]

results = []

results.append(
    evaluate_model(
        "Linear Regression",
        y_test,
        lr_pred
    )
)

results.append(
    evaluate_model(
        "Random Forest",
        y_test,
        rf_pred
    )
)

results_df = pd.DataFrame(
    results,
    columns=["Model", "MAE", "MSE", "RMSE", "R2"]
)

display(results_df.round(4))
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

Variable	Correlation with AQI
NO2	0.0111
Xylene	0.0104
Toluene	0.0094
NH3	0.0081
SO2	0.0060
CO	0.0029
NO	0.0019
PM10	0.0012
PM2.5	0.0007
O3	0.0006
Benzene	-0.0013
NOx	-0.0027

Table: Detailed Correlation Results